

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Third Semester B.E. (AIML) (Credit System) Degree Examinations
April 2023

21AM304 – INTRODUCTION TO MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Discuss the General approach to solve classification problem with an example. | 8 | L*2 | 1 | 1 |
| b) Illustrate the data aggregation used in data preprocessing and state the advantages and dis-advantages of it. | 6 | L2 | 1 | 1 |
| c) Explain different types of feature weighting techniques. | 6 | L2 | 2 | 1 |
| 2. a) With an example show how Hunt's algorithm can be used for constructing Decision Tree. | 10 | L2 | 1 | 1 |
| b) With a neat diagram, explain feature subset selection methodology. | 10 | L2 | 2 | 1 |
| 3. a) Define attributes. With an example illustrate the different types of attributes. | 10 | L2 | 2 | 1 |
| b) Write the algorithm for KNN and predict the following test data based on KNN. | | | | |

Test Data (T): **<Acid Durability=4, Strength=8, Class=?>**

Name	Acid Durability	Strength	Class
Type 1	4	3	Good
Type 2	6	5	Bad
Type 3	3	7	Bad
Type 4	2	8	Good

10 L3 2 2

Unit – II

4. a) For the given proximity matrix, construct dendrogram and nested cluster representation using single link (MIN) agglomerative hierarchical clustering (AHC) method.

	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

10 L3 3 2

- b) Write K-means algorithm and solve the following problem.
Apply K-Means algorithm to create 2 clusters by assigning each point to the nearest centroid for the given one dimensional point: {6, 12, 18, 24, 30, 42, 48}. The initial centroids can be chosen as {12, 42}.

10 L3 3 2

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|---|----|----|---|---|
| 5. a) Explain DBSCAN algorithm with an example. | 10 | L2 | 3 | 2 |
| b) Describe linear regression and multi-linear regression. | 10 | L2 | 4 | 1 |
| 6. a) With an example show the use of naïve bayes classification. | 10 | L2 | 4 | 1 |
| b) Discuss the how to fix hyperplanes in SVM. | 10 | L2 | 4 | 1 |

Unit – III

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|--|----|----|---|---|
| 7. a) Show the working of random forest algorithm. Explain the applications of random forest algorithm in brief. | 10 | L2 | 5 | 1 |
| b) Compare the bagging and boosting. | 10 | L2 | 5 | 1 |
| 8. a) Discuss how to reduce dimensionality using factor analysis. | 10 | L2 | 5 | 1 |
| b) Explain different method of solving imbalance problem. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome